

Performance Assessment of DECT-2020 NR and Classic DECT Coexistence Mechanisms

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Abstract—The recently standardized ETSI DECT-2020 New Radio (NR) technology promises to enable operator-independent Internet-of-Things (IoT) services. One of the supported operating bands is the 1880 – 1900 MHz band where IoT devices use one or more 1.728 MHz wide channels for multi-hop communications. It has been shown that such systems scale well to satisfy the requirements of 5G massive machine-type communications (mMTC). During the standardization process, the coexistence with classic DECT technology was addressed but the benefits of different options have not been deeply explored. In this paper, by utilizing system-level simulation techniques we evaluate several coexistence solutions for DECT-2020 and classic DECT systems including conventional listen-before-talk access (LBT), last-minute scan, and scheduling-based mechanisms. Our numerical results illustrate that the standard LBT operation ensures excellent performance with no more than 2% of packet drops for classic DECT, and still results in satisfactory DECT-2020 operation under 25% of resources allocated to classic DECT. The use of last-minute-scan leads to drastic performance degradation of DECT-2020 (by more than 10% in terms of packet drops) as compared to the standard LBT-only operation. Finally, a scheduling-based mechanism allows to improve the LBT performance for DECT-2020 devices by 2-5%.

Index Terms—DECT-2020; New Radio; IoT; coexistence.

I. INTRODUCTION

The amount of services offered by the Internet of Things (IoT) is growing each year spanning all the commercial sectors [1]. To enable connectivity for constantly growing IoT equipment massive machine-type communications (mMTC) technologies are being specified as a part of 5G IMT-2020. NB-IoT and LTE-M are the main mMTC technologies developed by 3GPP to provide cellular IoT (CIoT) connectivity with the former satisfying 5G IMT-2020 requirements [2].

In addition to 3GPP activities, there is an alternative ETSI initiative standardizing non-CIoT 5G mMTC technology. These efforts recently commenced in ETSI DECT-2020 New Radio (NR) standard in 2021 [3]. Current specifications define 19 different operating bands for DECT-2020 NR below 6 GHz, including unlicensed DECT bands [4]. The use of this 1880–1900 MHz band in Europe and Australia or 1900–1920 MHz band in other regions by utilizing eleven 1.728 MHz wide channels can be a game-changing opportunity for many IoT applications. This technology has recently been shown to provide mMTC device density of up to 4M/km² with 0.05%

packet loss rate (PLR) and 99% of packets experiencing delays less than 10 s satisfying 5G IMT-2020 requirements for mMTC services [5], [6]. Similarly to 3GPP New Radio (NR), DECT-2020 NR utilizes advanced physical and data-layer techniques such as flexible numerologies, HARQ, and listen-before-talk (LBT) mechanisms but inherently builds upon multi-hop tree-based topologies to deliver packets from the source mMTC device to the sink.

As the main operational bands for DECT-2020 NR are the DECT bands, the system design contains several features to have strong coexistence with classic DECT systems. Specifically, DECT-2020 NR frame and slot lengths, 10 ms and 416, 6 μ s, and one of the supported channel bandwidths of 1.728 MHz are exactly the same as in the classic DECT. However, as classic DECT was designed to support mainly voice calls, DECT-2020 NR is designed for a wider variety of services, resulting complete re-design of the channel access method of the DECT-2020 NR.

The LBT is the primary method for random access transmission to ensure coexistence with classic DECT or the other systems [7], [8]. The standard considers additional rules to detect periodic classic DECT transmissions that can be implemented to further improve coexistence. The question of coexistence is further complicated by the fact that the existing classic DECT technology does not utilize channel coding and existing equipment still relies upon the old generation voice codecs that are highly sensitive to packet losses. To enable smooth DECT-2020 NR deployments and high spectrum utilization of existing DECT bands, the coexistence between DECT-2020 NR and classic DECT technologies needs careful evaluation.

In this paper, we investigate the performance of coexistence between DECT-2020 NR and classic DECT technologies. Specifically, we assess the performance of the DECT-2020 NR mMTC system utilizing its base numerology and bandwidth option of 1.728 MHz, mesh network architecture, and random access transmission with LBT functionality. Then we consider various enhancements allowed for DECT-2020 including different variants of the “last-minute-scan” approach and scheduled-based mechanism. The main metric of interest is the PLR experienced by both technologies.

The main contributions of the paper are:

- detailed system level simulation campaign investigating

the performance of several DECT-2020 and classic DECT coexistence mechanisms;

- numerical results showing that LBT-based enhancements such as last-minute-scan operation drastically reduce DECT-2020 NR performance at no significant improvement for classic DECT;
- observation that the use of scheduled resource allocation is useful mainly for improving DECT-2020 NR performance in high load conditions.

Our paper is organized as follows. First, in Section II we describe the operation of DECT-2020 NR technology. Then, in Section III we introduce the considered coexistence mechanisms. The coexistence scenario and system parameters are introduced in Section IV. We report our numerical results in Section V. Conclusions are drawn in the last section.

II. DECT-2020 NR TECHNOLOGY

The DECT-2020 NR physical layer has been defined to support Time Division Duplex (TDD) frequency bands below 6 GHz. The physical layer employs Cyclic Prefix Orthogonal Frequency Division Multiplexing (CP-OFDM) combined with Time Division Multiple Access (TDMA) and Frequency Division Multiple Access (FDMA). The physical layer can support multiple numerologies, with different subcarrier spacing (SCS) options from 27 kHz up to 216 kHz, with appropriate Cyclic Prefix (CP) lengths. The used fast Fourier transform (FFT) sizes scale as well from 64 up to 1024 subcarriers, and when combined with SCS this allows operating with different transmission channel bandwidths, symbol lengths, number of symbols in slot, with the minimum transmission lengths of 5 symbols.

DECT-2020 NR transmission channel bandwidth can be from 1.728 MHz up to 221 MHz, whereas symbol duration can vary from 41.667 μ s to 5.208 μ s with the corresponding minimum transmission length of 208.3 μ s to small as 26 μ s. The physical layer utilizes Turbo channel coding for both physical control and physical data channels, denoted as PCC and PDC respectively. Transmission errors occurring for MAC PDUs carried in PDC are corrected by Hybrid ARQ with incremental redundancy allowing re-transmitting MAC PDUs and performing soft-combining between transmission and re-transmission in the receiver.

To enable a single design for all radio devices, disregarding their operational role in the system, as well as to enable reception of the different transmissions without prior information such as symbol synchronization, system broadcast, or ID of the transmitter, DECT-2020 NR defines a single transmission packet structure at physical layer. A transmission packet starts with a synchronization timing field (STF), followed by PCC and PDC. PCC supports two different payload sizes and four different demodulation reference signal (DMRS) patterns supporting up to 8 transmit antennas. After packet synchronization, it is required that the receiver is able to detect the used PCC payload size as well as the used DMRS pattern by the means of blind decoding. The number of decoding

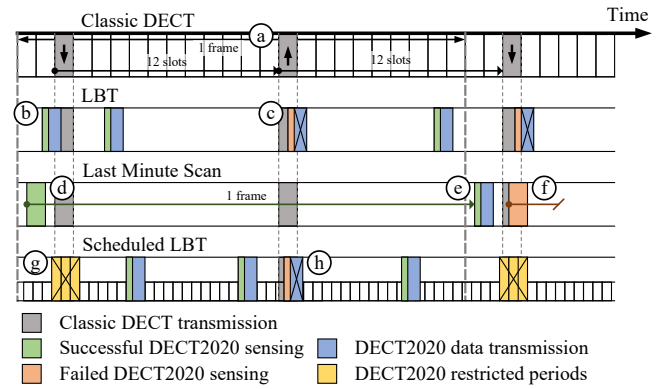


Fig. 1. DECT-2020 random access mechanisms on a single channel.

attempts is sufficiently modest and the receiver may optimize detection order freely.

In DECT-2020 NR mesh network architecture relies upon tree formations. In mesh architecture devices can communicate directly with each other, extending the network range and increasing communication reliability by operating at high signal-to-noise ratio (SNR) conditions and providing multiple alternative paths for the data. Due to the designed physical layer packet structure, the mode of radio devices may change depending on the communication context. A device may operate in so-called FT mode controlling radio resources of a cluster and receiving transmissions from its cluster member devices and initiate operation in so-called portable termination (PT) mode as a cluster member when forwarding received data and its own data to next device in next cluster three topology towards external systems for example. In mesh topology any device of the system may initiate operation in FT mode to allow other devices to associate with it. In that operational mode, device selects operating channel, performs transmission of network and cluster beacon messages as well as announces random access channel (RACH) resources, where other devices may initiate communication with the device.

Each device in PT mode select most optimum device(s) for association based on received link quality and signalled cost indicating how good path there is towards the external backend systems. Each device makes own autonomous routing decisions based for each routing PDU to which device the corresponding PDU should be transmitted. Thus, device being the source of the data, does not need establish and maintain data path before initiating communication, rather connection only to next hop is needed. This, together with the fact that devices are identified in single network with 32 bit MAC address allows DECT-2020 NR to support very high number of devices and with high device densities. In addition, that each routing device may select own operating channels DECT-2020 NR allows different routing devices share their cluster beacon message timing enabling, over the air synchronization between different cluster to optimize local interference conditions.

III. CONSIDERED COEXISTENCE MECHANISMS

In this section, we first introduce the classic DECT channel access mechanism and then proceed to specify the considered coexistence mechanisms for DECT-2020.

A. Classic DECT Channel Access

According to [9], classic DECT supports multiple frequency channels in different bands and utilized time-division multiple access (TDMA) for channel access. Each channel is divided into 10 ms frames and the frame consists of 24 slots. Prior to the first transmission classic DECT devices have to select a physical channel, which will be utilized throughout the whole connection duration unless any interference is encountered. Physical channel is a time-division duplex (TDD) pair, i.e. two time slots use the same channel and are half frame apart.

To select an appropriate slot pair, the classic DECT device first senses all possible slots in different frequency channels, orders them according to the measured receiver signal strength indicators (RSSI), and then selects the best slot pair on a single frequency channel. This procedure establishes a pair of recurrent time slots on a given frequency channel, Fig. 1(a). The selected slot pair is only allowed to be changed as a result of high PLR. If this happens, the described procedure is repeated. With this in mind, DECT-2020 random channel access should provide the means for detecting recurring transmissions in order to minimize its interference with classic DECT.

B. DECT-2020 LBT-Based MAC

The default DECT-2020 MAC configuration shall support the LBT with binary exponential back-off (BEB) mechanism for RACH transmissions [8]. According to this channel contention procedure, the radio device (RD) must perform channel sensing before the actual data transmission in order to avoid collisions. The sensing window is only two symbols out of the full slot located at the beginning of each slot. This may lead to situations when data transmission after successful channel access will overlap with recurrent classic DECT transmissions as depicted in Fig. 1(b). Recall that the classic DECT reacts to the packet losses instead of explicitly employing the LBT-like procedure. Accordingly, if RD senses any interference above the preset threshold, no data will be transmitted while entering the back-off state, Fig. 1(c).

C. Last-Minute-Scan Extensions

As classic DECT transmissions are recurrent in nature, i.e., another data transmission is assumed to happen exactly one frame after the previous one, the last-minute-scan was proposed as one of the mechanisms that enable DECT-2020 to detect this type of regular transmissions. The idea is to sense the channel for a full slot as compared to only two symbols sensing for the regular LBT procedure, making sure that this specific time slot is not occupied by an ongoing classic DECT transmission, see Fig 1(d). Otherwise, DECT-2020 RD will have to backoff and wait for another full slot opportunity, see Fig 1(f). One frame after that, Fig 1(e), under the assumption that there are no interfering classic DECT transmissions,

DECT-2020 RD performs a regular LBT procedure in order to validate that there are no other contenders for the channel. At that, RD is allowed to transmit data only if both sensing windows detect no interference.

This procedure can be implemented for data packets only or for both data and acknowledgement (ACK) packets. In what follows, we will evaluate both options. We refer to these options as basic and enhanced last-minute-scan, respectively.

D. Schedule-Based Extension

According to [8], DECT-2020 also supports the scheduled channel access. Data transfer in this case is performed by using pre-allocated resources. In this operational mode, the FT assigns the resources for RD to transmit and receive, thus, enabling the possibility to restrict RD random channel access if FT detects any recurrent interference in specific time slots, Fig 1(g). This approach may partially suppress interference leading to packet losses. However, there are still hidden node cases or situations when the FT detection threshold is too high, which still may result in failed receptions, see Fig. 1(h).

IV. THE CONSIDERED SCENARIO AND METRICS

In this section, we first define our coexistence evaluation scenario and then proceed with DECT-2020 and classic DECT configurations. Finally, we define the metrics of interest.

A. Evaluation Scenario

The original DECT spectrum specified 10 non-overlapped 1.728 MHz channels in the 1880 – 1900 band. Even though both systems have access to all of them, we consider the worst-case scenario, where DECT-2020 and classic DECT utilize the same channel under variable deployment and load conditions.

The second critical parameter is the device deployment and system load. For this there might not be a single solution, however, ideally, deployments should be generic presenting the overall situation, not focusing on a single type of deployment and not only the worst-case situation. Further, system loads should be variable increasing from low load to higher load to assess the impact of interference.

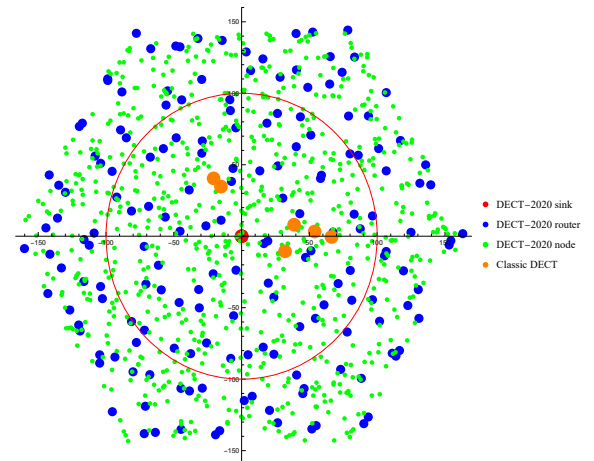


Fig. 2. Illustration of the considered deployment.

When considering the co-existence between the DECT-2020 NR mMTC solution, with a high number of devices and mainly utilizing RACH transmission, and the classic DECT system, one option is to utilize the ITU-R mMTC evaluation scenario defined in ITU-R M.2412 [10]. In that scenario, 80% of the devices are indoors and 20% of devices are outdoors with different channel models for RD to RD links and RD to FT links. However, the density of the DECT-2020 nodes should be lower than the ITU-R scenario which was intended to evaluate the maximum system capacity using multiple channels with 5 MHz of total bandwidth. The classic DECT connections should be randomly located inside a given area. The locations of the classic RDs are not the same as DECT-2020 NR FT as the two systems target different use cases.

Fig. 2 shows the considered scenario while Table I provides simulation parameters. The scenario is defined such that a single DECT-2020 NR FT (an FT with a backend connection operating as a sink) is in the center of the simulation area. The area is a circle with a radius of 170 m which roughly presents a situation where DECT-2020 FT is deployed with 500 m ISD as required by ITU-R. The data generated by the DECT-2020 system is transferred to FT in a multi-hop manner.

Multiple DECT-2020 RDs are randomly distributed in the considered area and their subset is assumed to operate in routing mode. Each RD operating in FT mode sends a beacon with a 2 second interval, which is used to form clustered tree routing topology towards the gateway FT. Each node generates application data which is transferred to gateway FT in a multi-hop way. Data generation frequency increases from one packet burst every 50 to 10 s, where each transmission contains 2 packets with 32 bytes of payload.

All the data transmissions are RACH transmissions with a length of one slot (416 μ s), and modulation and coding scheme (MCS) is selected in such a manner that MCS 1 can transfer 1 packet, MCS 3 is used for 2 packets, and MCS 4 – for 3 packets. The utilized MCS is selected based on link quality and the available number of packets in memory. All the DECT-2020 NR transmissions use constant power without power control. LBT is used to detect other transmissions and RD in FT maintains RACH resources constantly available — thus not avoiding time moments, where interference is detected.

A number of classic DECT pairs are dropped into a simulation environment at random locations within the circle of 75 m from the DECT-2020 sink. Each pair is then randomly and uniformly distributed within the circle of 20 m. In such a simulation scenario with Tx power of 10 dBm packet losses can be significant. Thus, classic DECT performance is provided based on the condition that the received signal strength is at least -85 dBm. In simulations, different classic DECT connections are infinite and considered ideal with respect to mutual interference between each other, i.e., different classic connections use randomly selected time slots that are different from each other and synchronized. No synchronization between DECT-2020 NR and classic DECT is assumed.

TABLE I
THE CONSIDERED SYSTEM PARAMETERS.

Parameter	Value
Simulation scenario parameters	
DECT-2020 NR FT	1, height 25 m, outdoors
Carrier frequency	1900 MHz
Number of channels	1
Channel bandwidth	1.728 MHz
Simulation area	radius of 170 m, 90800 m ²
Number of DECT-2020 NR RD	1000
DECT-2020 NR RD deployment	Uniformly random
Number of classic DECT devices	2 – 8 (1 – 4 connections)
Classic DECT deployment	radius of 75
Max distance between classic DECT	20 m
Building penetration loss	20 dB (ITU-R)
Indoor/outdoor fraction	80% / 20%
Channel model, Sink-Node	Urban macro
Channel model, Node-Node	Urban street canyon
Number of node/sink antenna elements	1
Sink/UE noise figure	7 dB
Sink/UE antenna element gain	0 dBi
Thermal noise level	-174 dBm/Hz
DECT-2020 NR parameters	
Traffic model	Bursts of 2 packets
Application data message size	32 bytes
Inter-arrival time	10-50 s
Channel access	LBT + extensions
Contention window size	min – 24, max – 768
LBT threshold	-75 dBm, -100 dBm
Minimum link quality	15 dB
Error correction	HARQ with 3 attempts
Sensitivity	-99.7 dBm (TS 103.636-2)
Tx power	10/19/23 dBm
Format	MSC 1,3,4 for 1,2,3 packets
Classic DECT parameters	
Traffic	1/1 UL/DL frames, 360 bits
Overall load	3 calls, 25% of system load
Interference	DECT-2020 only
Tx power	10 dBm
Sensitivity	-96 dBm
Target BER/PER at -96dBm	0.1% BER, 30% PER

B. Metrics of Interest

For a generic mMTC data service operating over DECT-2020 mesh system, the overall packet throughput and end-to-end PLR are often considered [10]. The overall end-to-end PLR should remain at a sufficiently low level so that if fully lossless data delivery is needed, end-to-end loss recovery operation would need to transmit a limited number of packets.

For classic DECT, when considering voice service, PLR is also considered as the main criterion. The PER should remain at such a level that after voice frame decoding, the quality of the connection remains acceptable for the user. By utilizing the results for PLR to mean opinion score (MoS) correspondence [11], for modern LC3plus codecs MOS is in the range of 3–4.5% PLR. However, however, with G.722 and G.726, for MOS to be over 3 PLR should be less than 2%.

V. NUMERICAL RESULTS

In this section, we analyze the coexistence performance of the baseline LBT mechanisms and those enhancements specified in Section III based on the scenario described in Section IV. The system parameters are specified in Table I.

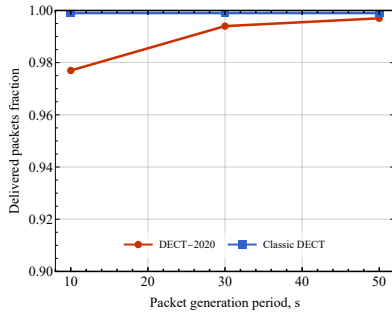


Fig. 3. DECT-2020 performance without classic DECT.

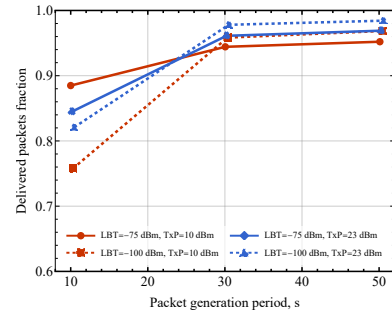
We start with the assessment of DECT-2020 and classic DECT technologies in isolation. In what follows, we will utilize this as a baseline. To this aim, Fig. 3 shows the delivered packets fraction for classic DECT and DECT-2020 technologies when operating alone at a single channel with 10 dBm Tx power when all transmissions are performed on RACH using LBT only. Note that the considered packet generation periods, 10, 30, and 50 s correspond to approximately 200, 75, and 45 packets per second (pps) at the sink. Analyzing the presented data, one may observe that when the packet generation period is less than 30 seconds, corresponding to 75 pps or less at the sink, the delivered packet fraction is virtually 100%. With a packet generation period of 10 seconds, the corresponding load is 200 pps per sink, and the delivered packet ratio is slightly lower than 100%, i.e., is around 98%. Classic DECT experiences virtually no packet losses.

A. Conventional LBT Operation

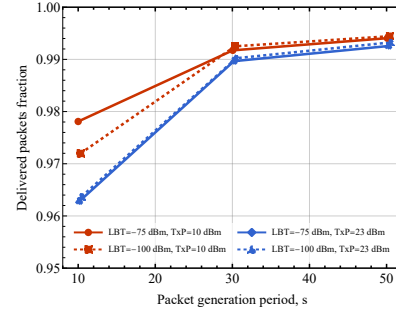
1) *Standard System Parameters:* We now proceed with the conventional LBT operation of DECT-2020. To this aim, in Fig. 4(a) we show DECT-2020 performance when 3 classic DECT calls are active simultaneously for two Tx powers and two LBT thresholds, and 10 dBm classic DECT power. Fig. 4(b) presents the performance of classic DECT when DECT-2020 NR uses LBT method only. Here, the classic DECT Tx power is kept constant at 10 dBm whereas DECT-2020 NR Tx power and LBT levels are modified.

Analyzing the result, one can note that when DECT-2020 RDs generate data packets every 30 seconds or lower, the DECT-2020 NR performance is almost the same as operating alone at the channel (98%). When operating with 23 dBm Tx power and the lowest LBT levels the performance is very marginally better than operating with 10 dBm and the highest LBT level. Similarly, classic DECT performance is very high at 99% for all the considered DECT-2020 NR configurations with less than 0.5% differences between each other.

Considering higher loads, e.g., inter-arrival period of 10 s (200 pps at the sink), one can notice that DECT-2020 performance goes down as compared to the case when operating alone, as the delivered packet fraction is between 80% to 90% as compared to 98%. Different Tx power levels have a somewhat higher impact than in the low load conditions. Further, a higher LBT level (-75 dBm) works clearly better



(a) DECT-2020



(b) Classic DECT

Fig. 4. DECT-2020 and classic DECT coexistence using LBT only.

than a very strict LBT level (-100 dBm), as with a -75 dBm level almost 90% packet delivery rate is achieved with 10 dBm Tx power. Nevertheless, at these loads DECT-2020 starts to be saturated and practical system operation starts to be unfeasible.

At the classic DECT side, the considered metric increases from 1% to 3% rate which starts to be at the edge of sufficient voice quality, especially with classic voice codecs [11]. However, the Tx power of the DECT-2020 does not introduce significant differences (1% at most) to classic DECT. Interestingly, a lower LBT level does not result in better coexistence than an LBT of -75 dBm, especially, with 10 dBm Tx power. This is due to higher system performance on the DECT-2020 side. Finally, we can observe that in all the cases the impact on the DECT-2020 side is significantly higher as compared to the impact on the classic DECT side.

2) *Increased Classic DECT Tx Power:* Fig. 5 presents the same analysis when classic DECT Tx power is increased to 23 dBm. It can be observed that when classic DECT Tx power is increased, the impact on DECT-2020 increases. With low loads the impact is modest but with high load conditions, the impact is more significant as the delivered packet fraction starts to be close to 70%. The increase of the impact occurs independently from DECT-2020 Tx power but with higher Tx power DECT-2020 NR operates slightly better. Additionally, the more stringent LBT level (-100 dBm) shows slightly better performance than the high LBT level (-75 dBm).

On the classic DECT side, the increase in Tx power does not introduce any significant difference. This is due to the fact that in the simulation minimum sensitivity level was set to -85 dBm resulting in some "unlucky" connections operating close to sensitivity level in "bad locations", where any interference

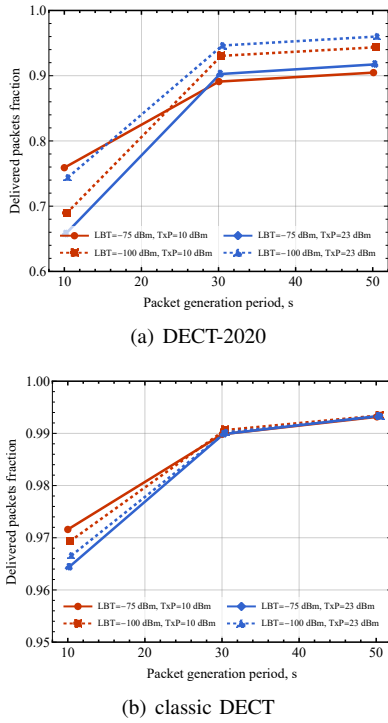


Fig. 5. The impact of increases DECT-2020 transmission power for DECT-2020 and classic DECT coexistence using LBT only.

or bad channel realization introduces packet losses. When the delivered packet ratios are above 96%, those connections dictate the achieved overall performance.

3) *Increased Received Power*: We now consider the case when the minimum received signal power of classic DECT is increased. This is achieved by reducing classic DECT RDs distance from each other to a maximum of 5 m instead of 20 m. This leads to the maximum path loss to be between 50 to 80 dB, which corresponds to received signal levels of -40 to -70 dBm with 10 dBm Tx power. The DECT-2020 NR system operates with LBT only with 10 dBm Tx power. This simulation scenario represents the situation, where classic DECT systems are used in a call center or in a restricted environment for high-quality audio systems. Fig. 6(a) indicates small improvement as compared to performance shown in Fig. 4(b). This can be explained by the fact that the overall area where classic DECT interferes with DECT-2020 is reduced.

Contrarily, Fig. 6(b) shows that the considered metric for classic DECT is clearly reduced. Classic DECT performance is high even in high DECT-2020 load conditions. It should be noted that simulating PLR below 1%, starts to be challenging from a system simulation point of view as simulation time for getting enough packet losses in each drop of devices will increase, increasing the total simulation time significantly.

B. Last-Minute-Scan Enhancements

1) *Basic "Last-Minute-Scan"*: Having studied LBT only performance of DECT-2020, we now proceed to assess the proposed enhancements. We start with the basic "last-minute-scan" operation. Fig. 6 shows classic DECT performance

when DECT-2020 NR performs the basic "last-minute-scan" for both data and ACK/NACK transmissions. It can be noted that with low load conditions (30 s inter-arrival period) there is no performance difference as compared to Fig. 4(b). Both LBT and LBT with basic "last-minute-scan" operation deliver roughly 99% packets. In high load conditions (10 s inter-arrival period), the "last-minute-scan" introduces roughly a 2% reduction to the classic DECT PLR, but performance is still lower than the performance shown in Fig. 4.

The reported marginal classic DECT gains come at a high cost for DECT-2020 NR system performance as seen in Fig. 6(a). Recalling Fig. 4(a), it can be observed that with "last-minute-scan" the PLR is decreased from 99% to 88% at low load (30 s inter-arrival period), and a staggering 66% at high load (10 s inter-arrival period). Thus, at low load, the performance is reduced by over 10% with no improvement to the classic DECT side, and at high load – by 15% with 2% PLR improvement at the classic DECT side.

2) *Enhanced "Last-Minute-Scan"*: We now proceed to present in Fig. 7 the enhanced version of the "last-minute-scan" implementation, where both data and ACK/NACK transmissions are subject to their own "last-minute-scan" process.

The impact on classic DECT with these schemes is identical to that shown in Fig. 6(b). However, it can be seen from Fig. 7, that if the Tx side needs to evaluate separately both data and ACK/NACK slot the performance is only slightly better in high load scenarios, DECT-2020 NR performance is still very low. Thus, we may conclude that the use of the "last-minute-scan" for DECT-2020 RACH should be avoided.

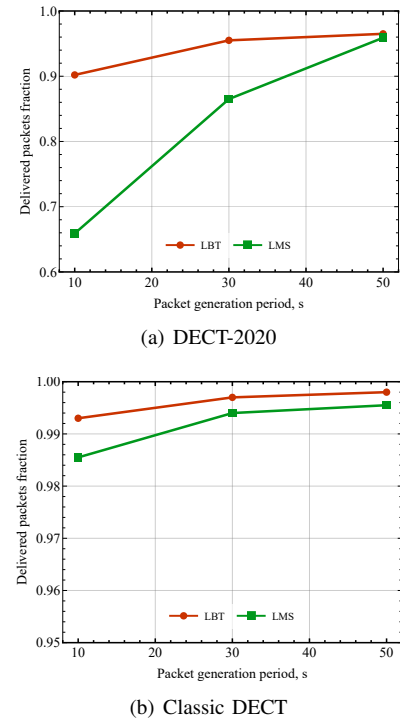


Fig. 6. Performance comparison of LBT and "last-minute-scan".

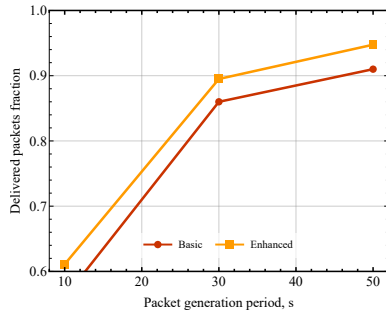


Fig. 7. DECT-2020 performance with enhanced "last-minute-scan" operation.

C. Scheduled Operation

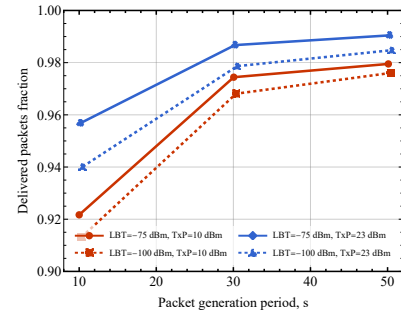
Scheduled operation is the last coexistence option that we evaluate. By analyzing the results presented in Fig. 8 and also comparing them to those in Fig. 4, we observe that packet delivery at high loads improves by almost 10%. In low loads, the difference is smaller as LBT only transmission can deliver data through also with high probability. Expectedly, this scheme still cannot replicate the performance when DECT-2020 NR operates alone on the channel as a significant part of the spectrum resources are used by another system.

Interestingly, the lower threshold of -100 dBm for detecting classic DECT and for LBT provides lower performance for DECT-2020 NR than the higher threshold of -75 dBm. This can be explained by the fact that with -100 dBm, the threshold level is too low and thus resources remain unused. Analyzing the classic DECT performance, we observe that roughly 1% improvement is obtained in high DECT-2020 NR load conditions as the packet delivery rate is improved from 97% to 98% independently of the utilized detection threshold and Tx power. The lower threshold of -100 dBm for detecting classic DECT and for LBT provides lower performance also for classic DECT than the higher threshold of -75 dBm. This is due to the overall lower spectrum utilization and efficiency at the DECT-2020 NR side. At low load conditions, there is no difference at all as the packet delivery ratio is above 99%.

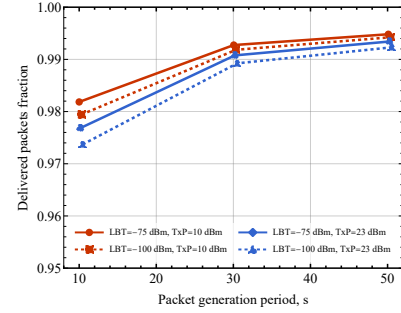
VI. CONCLUSIONS

In this study, we evaluated several coexistence mechanisms between DECT-2020 NR and the classic DECT. The considered enhancements can be implemented in the DECT-2020 NR system without modifications to the current standards.

Our results showed that the "last-minute-scan" operation reduces DECT-2020 performance as compared to the classic LBT mechanism. Therefore, instead of discussing more stringed requirements of "last-minute-scan" in DECT-2020 NR RACH transmission, one may consider removing it from RACH and rely only on LBT and FT's capabilities to allocate RACH resources. The use of scheduled access shows slight improvements over the conventional LBT mechanism. The gains approach multiple percent in terms of the DECT-2020 NR delivered packet fraction at no degradation to the classic DECT system. This extension can be recommended for improving the coexistence between the two considered systems.



(a) DECT-2020



(b) Classic DECT

Fig. 8. DECT-2020 and classic DECT coexistence using scheduled LBT.

However, the use of sophisticated RACH resource allocation is useful mainly for improving DECT-2020 NR performance in high load conditions as it provides some additional safety margin towards classic DECT.

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